

BMED 712 - Track A | Week 3 Report

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Period: Week 3 (post-submission - revision, debugging, retraining)

1. Summary of Completed Tasks

#	Task	Status
1	Address all 6 professor feedback items	Done
2	Identify and document feature extraction bug	Done
3	Validate corrected frequency-sheet features (300k windows)	Done
4	Retrain 3-class models (SVM / XGBoost / RF)	Done
5	First-ever 8-class (subtype) classification	Done
6	Sensor ablation with new features	Done
7	Phase-specific asymmetry analysis	Done
8	VGA-IMU ordinal correlation analysis	Done
9	Expanded feature experiment (proxy features)	Done
10	Revised Progress Report PDF	Done

2. Feature Extraction Bug - Root Cause Analysis

Bug 1 - Missing Acc channel

The extraction pipeline iterated over FreeAcc and Gyr signals only, **silently skipping the raw Acc channel**. Each of the 4 IMUs provides 3 signal types (Acc, FreeAcc, Gyr) x 3 axes, so 1/3 of available signals were never processed. The resulting file had 168 features instead of 216+.

Bug 2 - Trial-level aggregation

The old code averaged all time windows within each trial into a single row (1,356 rows = one per trial), collapsing within-trial temporal variability - a critical source of discriminative information for gait classification.

Bug 3 - No subtype label

The original CSV had only a 3-class label (Healthy / Neuro / Ortho). The 8-class cohort column (HS / PD / CVA / RIL / CIPN / KOA / HOA / ACL) was never written, making subtype analysis impossible.

Resolution

Teammate Fatemah re-extracted features from scratch: all 3 channels included, window-level rows (300,991 windows), both 3-class and 8-class labels, 216 features per window (4 sensors x 3 channels x 3 axes x 6 feature types). Missing-value rate: < 0.12%.

3. ML Results - Corrected Features

3.1 Three-class classification

Phase	Window	Overlap	Best Model	BAcc	F1	vs Old
post_turn	5 s	50%	XGBoost	79.2%	80.4%	+7.6%
pre_turn	5 s	50%	XGBoost	76.7%	76.2%	+5.1%
full_gait	3 s	50%	XGBoost	76.1%	75.9%	+4.5%
full_gait	5 s	50%	SVM	75.8%	75.0%	+4.2%
turn	1 s	50%	SVM	76.5%	75.3%	+4.9%

Old best (buggy features): BAcc = 71.6%.

3.2 Eight-class classification (subtype-level) - NEW

Phase	Model	BAcc	F1	Note
full_gait 5s/50%	SVM	41.5%	35.9%	Chance = 12.5%
pre_turn 5s/50%	SVM	41.0%	36.9%	
post_turn 6s/50%	SVM	39.9%	35.2%	

BAcc of 41.5% is 3.3x chance level, confirming IMU features carry subtype-discriminative signal.

3.3 Sensor ablation (full_gait 5s/50%, 3-class)

Sensor Set	SVM	XGBoost	RF
All (HE+LB+LF+RF)	75.8%	75.7%	75.3%
HE+LB	72.9%	75.1%	73.5%
HE only	72.7%	74.0%	73.2%
Feet (LF+RF)	70.4%	69.5%	67.1%
RF only	67.5%	67.9%	63.7%
LB only	71.0%	69.4%	64.8%

Key surprise: HE (head) sensor alone achieves 74.0% BAcc with XGBoost - retaining 97.8% of full-sensor performance.

3.4 Expanded feature experiment

Added 5 derived features per sensor-channel-axis (energy, DC ratio, relative variability, spectral complexity, normalized spectral power): 216 to 396 features. **Result: no improvement** (SVM 73.9% vs 75.8% original). Proxy features are correlated with existing stats. Raw signal access is needed for genuinely new features (kurtosis, ZCR, wavelet).

4. Professor's Feedback - All 6 Items Addressed

#	Feedback	Fix Applied
1	Tone down language	"potential indicators", "suggests"
2	Streamline narrative	Abstract: 2 contributions; Discussion: 3 paragraphs
3	ML improvement small	AUC labeled "modest"; clinical insight emphasized
4	Sensor ablation key-takeaway	Key-finding box: foot-only retains 93%
5	Fig 7: remove regression line	Spearman rho annotated; per-VGA boxplots
6	Table II: signed Cohen's d	All d negative: RIL -0.87, PD -0.77, ...

5. Key Insights This Week

- Feature extraction matters more than model tuning.** Fixing the missing Acc channel and using window-level data improved BAcc by +7.6% - more than any hyperparameter search could achieve.
- Head sensor is underrated.** HE alone (74.0%) nearly matches all 4 sensors (75.7%). A single head-mounted IMU could be a practical clinical screening device.
- Subtype discrimination is feasible.** 8-class BAcc of 41.5% (vs 12.5% chance) shows IMU features carry subtype-specific signatures.
- Straight-line walking carries the signal.** Pre-U-turn phase shows significant group differences ($p = 0.026$); yet ML performs best on post_turn (79.2%) - the return corridor has cleaner signal.
- VGA is a coarse proxy.** Spearman rho = -0.206 means VGA explains only ~4% of IMU asymmetry variance. IMU captures information invisible to clinical visual assessment.

6. Proposed Next Steps (Week 4)

- Access raw IMU signals to compute genuinely new features (kurtosis, skewness, zero-crossing rate, wavelet energy)
- Re-run 8-class with SMOTE or class-weighted loss to address RIL/ACL imbalance
- Phase-feature fusion: concatenate pre_turn + post_turn features per trial

- Discuss with professor: is head-sensor-only IMU a publishable finding?
 - Begin final report structure planning
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7. Files Delivered This Week

File	Description
analysis/validate_new_features.py	Feature validation (300k windows)
analysis/train_new_features.py	3-class, 8-class, sensor ablation
analysis/expand_features.py	Expanded feature experiment
analysis/fix_fig7_table2.py	Corrected Fig 7 + signed Table II
results/ml_new_features/	All ML result CSVs and plots
results/validation/	Validation outputs (heatmaps, KDE)
results/Progress_Report_Revised.pdf	Revised report (all feedback)

Week 3 Report (Chinese)

一、核心发现：特征提取 Bug 分析

旧版 master_features.csv 存在两个复合错误：

错误1 - Acc 通道缺失：旧代码只处理了 FreeAcc 和 Gyr，完全跳过了原始 Acc 通道。每个 IMU 有 3 种信号 x 3 轴，旧版丢失了 1/3 的信号源 (168 vs 216 特征)。

错误2 - 聚合粒度错误：旧代码将每个 trial 内窗口取均值 (1,356 行)，丢失了 trial 内时间变异性。

错误3 - 无亚型标签：只有 3 类标签，无 8 类 cohort 标签。

修复：Fatemah 重新提取全部特征：300,991 个窗口，3 通道 + 8 类标签，缺失率 < 0.12%。

二、修正后 ML 结果

Experiment	Best Model	BAcc	vs Old
3-class (post_turn 5s/50%)	XGBoost	79.2%	+7.6%
3-class (full_gait 5s/50%)	SVM	75.8%	+4.2%
8-class (full_gait 5s/50%)	SVM	41.5%	New (chance 12.5%)

传感器消融新发现：头部 (HE) 单传感器 XGBoost 达到 74.0% BAcc，接近全部 4 传感器 (75.7%)。扩展特征实验无提升 (396 vs 216 特征)。

三、导师反馈 - 全部 6 条已回应

- 语气软化 ("robust biomarkers" -> "potential indicators")
- 叙事精简 (摘要重写，贡献前置)
- ML 定位调整 (AUC 0.716 标注 "modest")
- 传感器消融关键结论 (foot-only 保留 93%)
- 图7修正 (删除回归线，改用 Spearman + 箱线图)
- 表II修正 (Cohen's d 全部改为负值)

四、本周关键洞见

- 特征工程 > 模型调参：修复 Acc 缺失 + 窗口级数据带来 +7.6% BAcc。
- 头部传感器被低估：HE 单传感器 (74.0%) 几乎匹配全部 4 传感器。
- 亚型分类可行：8 类 BAcc 41.5% (随机 12.5%) 证明 IMU 携带亚型信号。
- 直线行走 = 主信号源：pre-U-turn $p=0.026$ 显著；ML 在 post-turn 最佳 (79.2%)。
- VGA 是粗糙代理：Spearman $\rho = -0.206$ ，仅解释 ~4% 方差。

五、下周计划

- 获取原始 IMU 信号，提取峰度/过零率/小波能量等新特征
- 8 类分类使用 SMOTE 或类别加权损失缓解不平衡
- 尝试 pre_turn + post_turn 阶段特征融合
- 与导师讨论：头部单传感器发现是否可作为可发表亮点
- 开始规划 final report 结构